

UI Section Guide

Stage 7 Streamlit Dashboard

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1 Purpose of This Document

This document explains what each section of the Stage 7 Streamlit UI is for, who it is intended for, and how each page should be interpreted. It is meant to help a marker, supervisor, or bank-style reviewer understand the dashboard quickly without having to infer the purpose of each page from the code.

Primary navigation is defined in `src/ui/app.py`. The sidebar pages are:

1. Research Overview
2. Regional Signals
3. Region Comparison
4. Scenario & Diagnostics
5. Household Explorer
6. Methodology
7. Technical Controls
8. Limitations

2 Research Overview

Purpose: provide the main top-level research read for the UK housing market and a single selected region.

Audience: general reviewer, supervisor, bank-style research reviewer, or first-time user.

What this page contains:

- House View Summary;
- Market Snapshot;
- compact validation-status summary;
- Key Housing Signals for the selected region;
- Risk Context via scenarios and simulation calibration;
- optional cross-regional orientation charts;
- coverage and source-detail tables.

How to interpret it:

- this is the best starting page for understanding the project;
- valuation and downside context should be read before any forecast-like use;

- later descriptive data are shown as context only and are not silently folded back into the calibrated model.

What this page is not:

- not a trading screen;
- not a real-time market dashboard;
- not a property-specific recommendation tool.

3 Regional Signals

Purpose: provide the main analyst-facing regional dashboard for institutional-style review of one region at a time.

Audience: research analyst, marker, dissertation reviewer, or bank-style model reviewer.

What this page contains:

- Headline View on valuation, cycle, and downside;
- Supporting Signals with evidence tags;
- Supporting Charts including fan chart, score breakdown, scenario comparison, and yield-versus-risk context;
- research-style commentary in plain language.

Core signals shown here:

- Valuation: strongest evidence base;
- Cycle: partially supported;
- Downside: scenario and risk context;
- Rent support / yield: contextual rather than fully validated.

How to interpret it:

- this is the core institutional page;
- the component signals matter more than the blended score;
- a reviewer should focus first on valuation, then downside, then cycle.

What this page is not:

- not a probability-weighted forecast table;
- not a REIT backtest engine.

4 Region Comparison

Purpose: compare a small set of regions over time on the same chart.

Audience: anyone trying to understand regional dispersion rather than a single-region case study.

User inputs:

- up to four regions;
- comparison metric;
- time window.

Metrics available:

- Nominal House Price Index;
- Real House Price Index;
- Fair Value Gap;
- Downside Probability;
- Rental Yield.

How to interpret it:

- this page is for relative comparison;
- it is useful for seeing whether London and the regions are moving together or diverging;
- when Fair Value Gap is selected, the page also shows the latest gap and confidence-interval context where available.

What this page is not:

- not the main decision page;
- not the place to judge overall model validation.

5 Scenario & Diagnostics

Purpose: stress-test a region under alternative macro assumptions and inspect the simulation behaviour.

Audience: analyst or technically engaged reviewer.

User inputs:

- region;

- scenario preset;
- additional rate shock;
- volatility multiplier;
- drift override;
- horizon;
- number of simulation paths.

What this page contains:

- scenario-specific fair-value and rate assumptions;
- fan chart under the selected stress setup;
- scenario comparison table across presets;
- downloadable scenario-path CSV;
- simulation calibration diagnostics.

How to interpret it:

- this is a perturbation and stress-testing tool;
- it does not re-estimate the econometric model live;
- it helps answer what happens if the macro path is worse, better, or more volatile than baseline.

What this page is not:

- not a model retraining page;
- not proof that a given macro scenario is the most likely one.

6 Household Explorer

Purpose: translate the research system into an affordability and downside view for a household-level example.

Audience: non-technical user, student reviewer, or someone wanting to see consumer-side implications.

User inputs:

- region;
- household income;

- deposit;
- mortgage term;
- target property price;
- risk tolerance;
- macro scenario.

What this page contains:

- plain-language headline view for the chosen household setup;
- supporting signals for affordability, valuation, cycle, and downside;
- monthly payment and mortgage stress outputs;
- fan chart and scenario comparison;
- key risks and caveats.

How to interpret it:

- this page is secondary to the core regional research dashboard;
- it is useful for illustrating affordability pressure and downside exposure;
- it should not be used as mortgage advice or as a buy / do-not-buy decision engine.

What this page is not:

- not financial advice;
- not lender underwriting.

7 Methodology

Purpose: explain how the model works, what data it uses, and what the current validation position is.

Audience: marker, technical reviewer, supervisor, or model-risk reviewer.

What this page contains:

- high-level explanation of the pipeline design;
- the distinction between model-panel data and descriptive data;
- current validation status;
- structural stability tests;

- signal backtest summaries;
- links to wider project documentation;
- glossary of terms.

How to interpret it:

- this is the main “how it works” page;
- it should be read together with the Limitations page;
- it is the best page for defending the project academically or institutionally.

8 Technical Controls

Purpose: expose calibrated parameters and override guardrails without allowing users to change the core model calibration.

Audience: technical reviewer or analyst.

What this page contains:

- region-level calibrated parameters such as `kappa`, `sigma`, `mu_equilibrium`, `P_star_now`, and `gamma_annual`;
- interactive guardrails for scenario overrides;
- glossary-backed parameter explanations.

How to interpret it:

- this page improves transparency and auditability;
- it shows what the model is actually using under the hood;
- the values are read-only and are not meant to be hand-tuned here.

What this page is not:

- not a calibration editor;
- not a research notebook.

9 Limitations

Purpose: state the model risks, validation weaknesses, and practical constraints explicitly.

Audience: everyone. This page is mandatory reading before serious use.

What this page contains:

- 4 material risks;
- 10 significant limitations;
- 8 technical notes;
- data-currency warning based on `effective_data_as_of`.

Key themes covered:

- publication lag and stale-data risk;
- uneven source coverage;
- holdout validation weakness versus zero-growth;
- regional calibration issues;
- descriptive versus validated signal boundaries;
- structural break and simulation limitations.

How to interpret it:

- this page defines the safe boundary of the project;
- if the Methodology page explains how the system works, this page explains where it can fail.

10 Recommended Reading Order

For a first-time reviewer:

1. Research Overview
2. Regional Signals
3. Methodology
4. Limitations
5. Scenario & Diagnostics

For a bank-style or dissertation-style assessment:

1. Methodology
2. Limitations
3. Research Overview
4. Regional Signals
5. Technical Controls

11 One-Sentence Summary

The UI is structured so that the user starts with a high-level regional housing research view, drills into region-level signals and scenarios, and then uses the Methodology and Limitations pages to understand exactly how much weight the outputs should carry.